1 Introduction

1.1 Definition of masting

- Observations of masting in relation to seed predation and flowering

1.2 Major hypotheses for masting

- Predation satiation
- Pollination coupling
- Resource matching

1.3 Hypotheses operating at different reproductive stages

- Predation satiation—seed maturation/dispersal stages
- Pollination coupling—flowering/pollination stages
- Resource matching—across stages
- Mechanisms are not mutually exclusive
- Traits thus might be important to understand masting

1.4 Trait-based perspective

- Literature review on understanding masting from traits perspective
- Briefly connect traits to mechanisms
- Not clear connection between traits and hypotheses

1.5 Introduce framework linking hypothesis x traits x masting

- Link traits to hypotheses help understand drivers of masting for different species
- Here we use North American tree data to test this framework
  - Masting info available
  - Includes both masting and non-masting species
  - Not using CV data because don't have stand-level traits data

2 Methods

2.1 Data collection

- Data scraping
  - Data sources: Silvics of North America, Kew Seed Dataset, TRY Dataset
  - Trait selection
  - Keywords used for data scraping
    * Definition of strong masting vs. non-masting (referred to as masting vs. non-masting moving forward)
    * Criteria for other traits
- Data cleaning
  - Cleaning of categorical traits
  - Decision for numeric traits with a range

2.2 Statistical analysis

- Main analysis for masting pattern and single trait: phyloglm
- Masting pattern with a multivariate analysis: NMDS
- Phylogenetic signal check for traits: Pagel's λ and D-statistics

3 Discussion

3.1 Summary of main findings

- Seed weight supports the predation satiation hypothesis
- Pollination supports the pollination coupling hypothesis
- Discuss mechanisms in the context of existing literature
- Other traits show some pattern but statistically non-significant
- Transition to alternative hypotheses

3.2 Masting is a population-level phenomenon

- Examples of species that mast only under certain environmental conditions
- Traits need to be measured locally

3.3 Different species selected under different hypotheses

- Introduce flowering masting vs. seed masting
- Different species driven by different mechanisms
- Need a more refined definition of masting and additional flowering-related traits
- Transition to phylogeny

3.4 Evolutionary scale

- Angiosperm vs. gymnosperm results
  - More masting gymnosperm sp
  - Longer leaf longevity in gymnosperm
  - More monoecious species in gymnosperm
  - Wind pollination is universal among gymnosperms
- Importance of testing within clades (*Quercus.*)
- Importance of looking for potential evolution support between gymnosperm and angiosperm